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Ultralong, spin-photon fibres enable polarization-enhanced wearable sensing

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Abstract

Photon-in-textile offers transformative potential for wearable sensing, yet persistent challenges in signal overlap, coupling, and quality degradation in dynamic, multivariate environments limit their efficacy. A new polarization-enhanced sensing technology, enabled by a spin fibre-textile capable of efficient decoupling between multivariable interference, is presented. We develop a discrete helix anchoring strategy that autonomously embeds circularly polarized materials within fibres during extrusion, yielding kilometer-scale spin-photon fibres—exceeding 1,300 m in continuous length, a scale unprecedented in prior reports. More importantly, the resulted core-sheath beaded fibres exhibit a luminescence asymmetry factor of 0.41, an improvement of one-order-of-magnitude over existing reports. The fibre-woven fabrics are then produced, allowing for dynamic, real-time signal acquisition by our developed polarization-enhanced sensing approach—that is, distinguishing spin light from surrounding optical fields—achieving signal segmentation and noise suppression at source with 92.63% signal entropy reduction in dynamic scenarios. This spin-photon-digital signal conversion system realizes a superior normalized signal-to-noise ratio of 1.0 (noiseless), thereby enabling multi-dimensional robotic control with 100% sensing accuracy under interference. Furthermore, we demonstrate the scalability and compatibility of this technology in polarization-based image processing, target recognition, and virtual reality. This work offers an innovative solution for robust, embedded intelligence in soft robotics and human-machine symbiosis.

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Introduction

The integration of photonic science and textile engineering promotes the transformative advancements in wireless wearable sensing systems[1–3], unlocking unprecedented opportunities for robotics[4–7], intelligent industries[5,8], and telemedicine[9]. Fabric-based photonic sensing is distinguished by its inherent flexibility, biocompatibility, and seamless integration potential with everyday apparel[3,5,10]. However, achieving reliable signal quality in dynamic, multivariate real-world environments remains a critical challenge[10–15]. The target signals frequently overlap and couple with diverse interferences (*e.g.*, ambient noise or irrelevant sources), resulting in signal drift or distortion (Supplementary Fig. 1). This leads to significant amplification of computational complexity and error susceptibility for valid data extraction[10,12,13]. Moreover, high-density sensor arrays also exacerbate signal crosstalk, further degrading data fidelity[16,17]. These barriers hinder precision perception and limit the scalability of photonic textiles in real-world robotics.

We, took a view, employed optical spin to address this issue. When spin photons carrying identical helicity ($+\hbar$ or $-\hbar$) undergo spatially localized aggregation within a specific constrained system, their collective behavior manifests macroscopic circularly polarized optical signatures[18,19]. Such special photons would offer high-dimensional optical information—distinct from intensity or wavelength-prioritizing precision—in interactive sensing for complex real-world environments[20]. Thus, encoding circular polarization states in fibres shows promising to efficiently decouple target signals from interference (Supplementary Fig. 2a), leading to substantially enhanced signal fidelity[21–23]. However, chiroptical fibres have been rarely reported to date, exhibiting unsatisfactory luminescence asymmetry factor ($g_{lum} \approx 10^{-3}$-10^{-2}) and restricted length—due to internal inhomogeneous chiral molecular alignment

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arising from conventional twist-spinning synthesis—conflicting with industrial manufacturing demands and thus hindering the practice. Achieving overall-control of strong circular-polarization, fine processibility and large-scale fabrication is still an unsettled but urgent dilemma.

Here, we proposed a new discrete helix anchoring strategy, in which continuously-fabricated independent "warehouses" ensured microscopic chirality and stability, settling long-range possibility. This fixed-point strategy mapped circularly-polarization to helical anisotropic fibre structures, achieving unprecedented performance—high g_{lum} up to 0.41 and kilometer-scale fibres length exceeding 1,300 m. Our designed fibres obtained a bending radius of 10^{-5} m, showing an advancement for wearable sensing compared to the rigid components in traditional circuitry (Supplementary Fig. 2b). Furthermore, our polarization-enhanced sensing approach leveraged the discriminative properties of spin photons against ambient optical fields to achieve real-time signal segmentation and noise suppression at source, reducing signal entropy by 92.63% in dynamic scenarios. Lastly, signal generation, decoupling, and transmission functionalities were integrated into a single fibre, circumventing multi-component interconnect losses and realizing an interference-suppressed normalized signal-to-noise ratio (NSNR = 1.0, noiseless) along with 100% accuracy of remote robotic arm sensing across 200 test cycles. This technology demonstrated an integration of polarization-based image processing, target recognition, and virtual reality (VR), promoting next-generation intelligent robotics.

Results and discussion

Spin-photon fibres enabled polarization-enhanced sensing

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We first present a holistic framework for the polarization-enhanced sensing with our designed spin fibre-woven-textile (Fig. 1). The stimulated fabrics emit spin-photonic signals, of which helicity information ($\pm\hbar$) serves as distinguishable "optical fingerprints" (Fig. 1a-i). These signals propagate through ambient optical fields while maintaining decoupled polarization states, and are subsequently captured by the dual-channel imaging system equipped with left- (L) and right- (R) handed circular polarizers (Fig. 1a-ii). The computational algorithms within the imaging system real-timely decode the photonic-spin signals via differential decomposition of the optical vector field, extract the effective circularly polarized components of the target, and map them as photonic-spin density images. This approach addresses the pervasive "signal mixing" issue in conventional imaging[11–14], where overlapping or irrelevant interference degrades recognition accuracy, by exploiting orthogonally decouplable polarization states (Supplementary Note 1). The separability of these specialized signals also leads to exceptional normalized signal-to-noise ratio (NSNR = 1, SNR = 82.04 dB, Supplementary Note 2) and markedly lower the information/signal entropy (Supplementary Note 3). This, in turn, ensures the robustness of the polarization-enhanced sensing (Fig. 1b-i). Building upon these, we apply the technology to machine vision tasks with superior accuracy compared to conventional image sensing (Fig. 1b-ii). Going one step further, we integrate this system into robotic teleoperation and virtual reality (VR, Fig. 1b-iii), achieving 100% accuracy in response—particularly crucial for precision operations (*e.g.*, microsurgery and semiconductor fabrication) tolerating no errors[24,25].

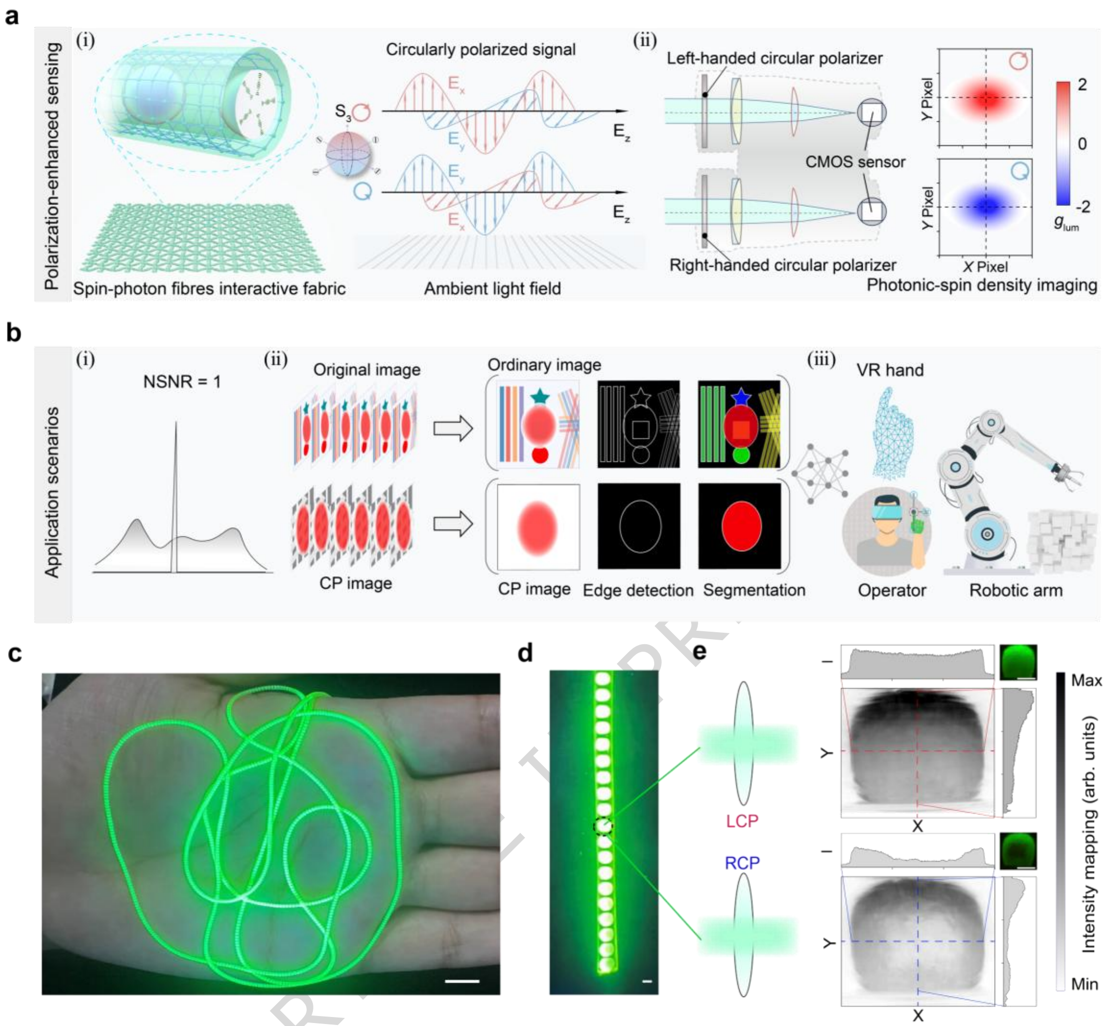

Fig. 1 | Spin-photon fibres enabled polarization-enhanced wearable sensing. a, Schematic illustrates the polarization-enhanced sensing based on designed fibre-fabric, involving photonic-spin signals generation (**i**), transmission, and decoding (**ii**). **b**, Illustration of polarization-enhanced sensing advantages (**i**), its working scenarios in machine vision tasks (**ii**), and extended applications in virtual reality (VR) and robotic teleoperation (**iii**). **c**, Photograph of a fibre on hand, showing its skin-friendness. Scale bar, 1 cm. **d-e**, Optical microscope image of helical microspheres within a fibre (**d**), generating photonic-spin signals with differential intensity under circular polarizers (**e**). Scale bars, 500 μm.

Moreover, our designed fibres exhibit satisfactory skin compatibility and mechanical flexibility for wearable integration (Fig. 1c). Each inner helical microsphere in fibres functions as an independent photonic-spin emitter—showing distinct light-dark intensity differences under circular polarizers (Fig. 1d,e).

Spin-photon fibres fabrication

To materialize the designed optical fibres and achieve large-scale synthesis, overcoming long-range stability and strong circular-polarization is indispensable but unachievable. This is due to difficult global-control of apposite compatibility, matching of microsphere of pitch, and consecutive processability. We developed a discrete helix anchoring strategy—embedding helical microsphere emitters within fibre through coaxial microfluidic spinning (Fig. 2a and Supplementary Fig. 3). We selected the cholesteric liquid crystal as the chiral host and the laser dye as the emissive component to construct the intensely circularly-polarized complex, which maintained the photostability of the dyes during the composite process (Supplementary Figs. 4-7).

Meanwhile, we chose gel polymer (sodium alginate) as "micro-warehouse" for polarization isolation in fibre, and its aqueous but viscous nature anchored ordered inner structure without floating (Supplementary Fig. 8). The generated flow shear force[6,26] stemmed from well-chosen flow rate ratio between two phase also contributed to fabrication of anchored-microsphere-in-fibre structure. The fibre structure was stabled by thermodynamic phase separation and mechanical fixation by the gel network during the process (Supplementary Fig. 9). After that, crosslinking degree of fibre increased untill solidified as ion exchanging in coagulation bath. Finally, its hydration stability enhanced through glycerol, which filled fibre gaps and prevented brittleness resulting from water loss (Supplementary Fig. 10). This method integrates signal emission and transmission functions into a single emitter, eliminating reliance on bulky, rigid optical components in wearable sensing, evidenced by phase-field modeling (Fig. 2b,c, Supplementary Fig. 11 and Supplementary Note 4).

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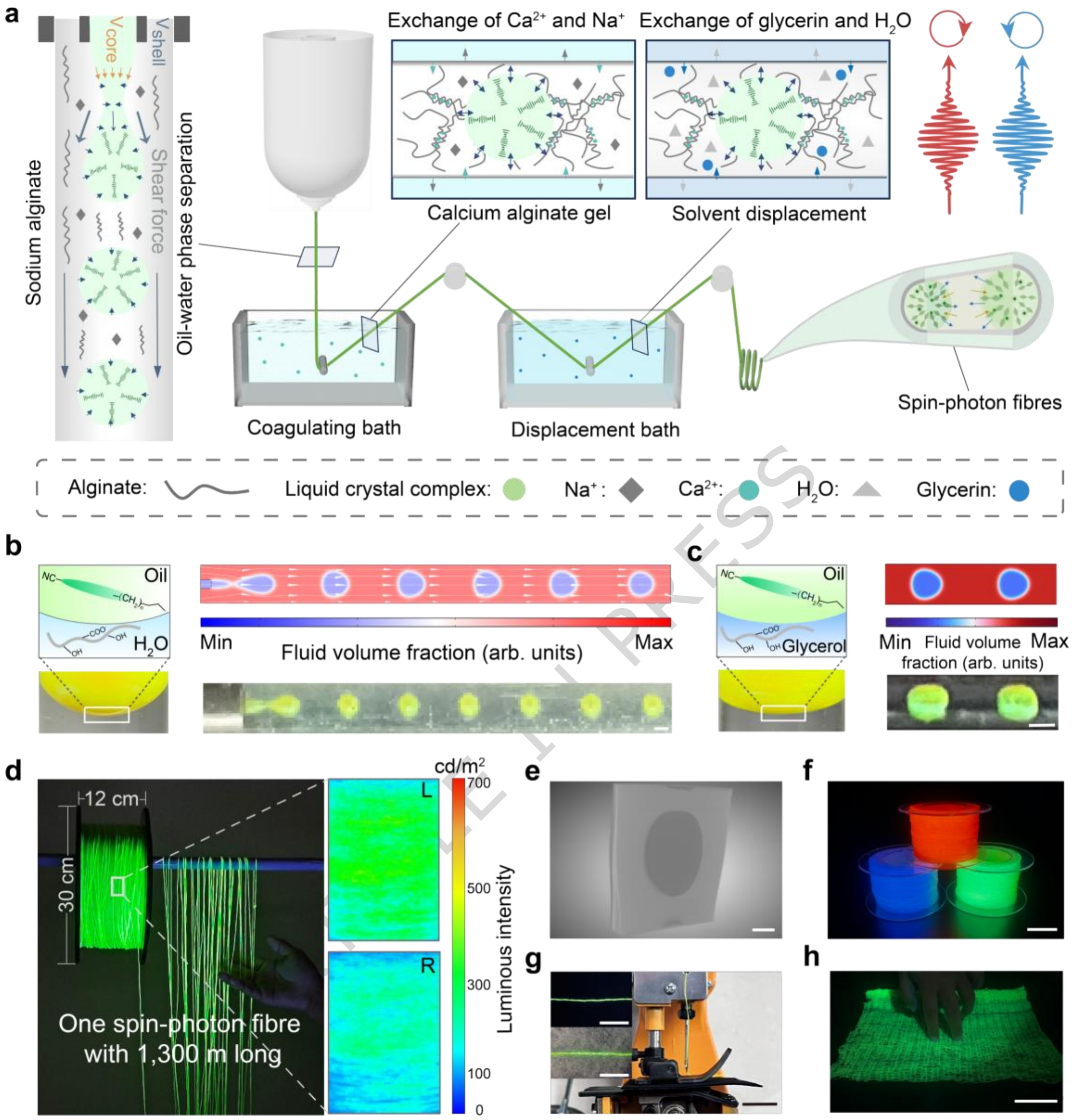

Fig. 2 | Spin-photon fibre fabrication. a, Schematic showing the setup used to continuously produce spin-photon fibres. **b-c**, COMSOL simulation (top) and photograph (bottom) of coaxial fluid shear and phase separation (**b**), and phase field after solvent exchange (**c**). Scale bars, 500 μm. **d**, Photograph of a single fibre with the length of 1,300 m and the corresponding intensity contrast under circular polarizers. **e**, X-ray three-dimensional reconstruction, revealing the structure of the embedded microsphere. Scale bar, 100 μm. **f**, Photograph of 300 m red, green and blue luminescent fibres. Scale bar, 5 cm. **g**, Photograph of the industrial embroidery processing of fibres. Scale bar, 1 cm. **h**, Photograph of a fibre-woven textile. Scale bar, 5 cm.

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As expected, we achieved a scalable production of the spin-photon fibres, obtaining an ultra-long fibre with the length up to 1,300 m (the longest polarized fibre to date) with considerable circular polarization (Fig. 2d). These fibres maintained favorable performance at multiple positions along fibre length and across batches, revealing their reproducibility and scalability for textile manufacturing (Supplementary Fig. 12). X-ray three-dimensional reconstruction proved the microscopic phase-separated structure (Fig. 2e). Going one step further, we fabricated 300-meter-long red-, green-, and blue-emissive fibres (Fig. 2f and Supplementary Fig. 13), respectively, meeting the multi-emission requirement of smart textile (Supplementary Fig. 14). The fabricated spin-photon fibres were soft and tough (bending radius of 10^{-5} m, Young's modulus of 4.17 MPa and an average fracture strength of 0.56 MPa), could undergo operations such as twisting and knotting (necessary for wearable applications), and support loads up to 500 g (Supplementary Figs. 15a-c and 16). They also remained stable in laundry detergent solution and cyclohexane (Supplementary Fig. 15d,e), satisfying the fibre industrial processing[5,8]. Moreover, we conducted photobleaching tests, and confirmed their long-term UV stability under continuous illumination (Supplementary Fig. 17). We then involved these fibres in industrial embroidery and interactive textile weaving, verifying their fine processing compatibility with modern textile manufacturing (Fig. 2g,h and Supplementary Fig. 18). We used a light strip as the external UV source, and confirmed its backside irradiance (to skin) was 0.035 mW/cm^2, lower than the safety limit (0.1 mW/cm^2, Supplementary Fig. 19).

Optical characteristic investigation

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We then investigated the circular polarization properties of synthesized fibres, of which photonic-spin emission originated from the spin-selectivity of chiral architectures to the embedded molecular emitters (Fig. 3a). When the inner dyes were photoexcited, they emitted photons carrying spin angular momentum (SAM)[22]. These photons traversed a one-dimensional photonic crystal structure formed by the cholesteric liquid crystals, undergoing spin-selective filtering—preferentially transmitting helicities ($\pm\hbar$) opposite to the host. When the helicity of the photons matched the chiral liquid crystal, constructive interference occurred, resulting in enhanced reflection (Supplementary Fig. 20 and Supplementary Note 5) and selective filtration of the photons[27]. The spin polarization rate P_S is the degree of collective spin orientation of photons[28,29]:

$$P_S = \frac{N_L - N_R}{N_L + N_R} \tag{1}$$

$$\langle S_z \rangle = -\hbar \cdot P_S \tag{2}$$

where N_L and N_R represent the number of left- and right-handed photons per unit volume, and $\langle S_z \rangle$ is the average spin angular momentum density of photons. The net helicity imbalance ($N_L \neq N_R$) among spin-photons generated a nonzero spin density[28], their collective behaviors (P_S) finally manifesting as macroscopic circularly polarized signatures. The mathematical expectation of spin angular momentum converges to $+\hbar$ or $-\hbar$ as the statistical probability of photons with specific helicity approaches 1[22,27], exhibiting fully circularly polarized optical signals (Supplementary Note 5).

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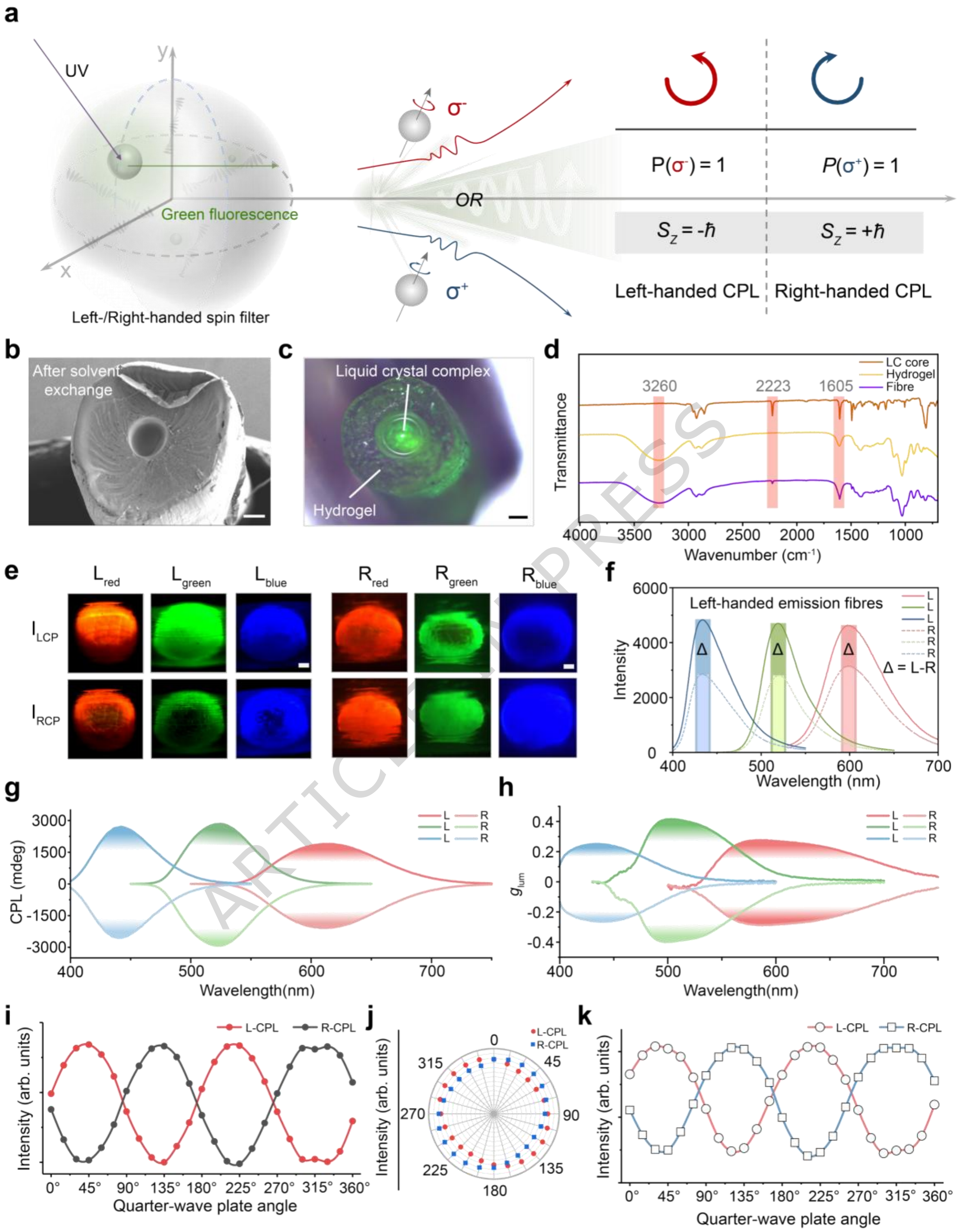

Fig. 3 | Circular polarization examination of fibres. a, Schematic of the photonic spin selectivity of the helical microsphere-in-fibre. **b-c**, SEM (**b**) and optical microscopy (**c**) images of the cross-section of a fibre after solvent exchange. Scale bars, 100 μm. **d**, FTIR spectra of the fibre and its precursors. **e-f**, Optical microscope images (**e**) and fluorescence intensity distribution (**f**) of red, green, and blue microspheres within fibres under circular polarizers. Scale bars, 10 μm. **g-h**, Circularly polarized luminescence (**g**) and the corresponding g_{lum} (**h**) spectra of red-, green-, and blue-emissive fibres. **i-j**, Emission intensities of left- and right-handed fibres as a function of polarization angle with (**i**) and without (**j**) quarter wavelength plate (in dark). **k**, Polarization angle (under 274 lux) effect on emission intensity of fibres, showing the environmental robustness and real-world applicability.

In our synthesized fibres, the robust photonic-spin emission was inseparable from stable helical microsphere structure. Scanning electron microscopy (SEM) and optical microscopy revealed the well-defined core-sheath architecture containing chiral liquid crystal within the fibre (Fig. 3b,c). Fourier-transform infrared (FTIR) spectroscopy also verified the independent structure without newly-formed chemical bond formation during fabrication, since both the chiral liquid crystal and hydrogel maintained their characteristic peaks—the cyano (2223 cm^{-1}), the hydroxyl (3260 cm^{-1}), and the carbonyl (1605 cm^{-1}) stretching vibrations (Fig. 3d). These separated and stabilized microspheres showed distinct red, green, and blue circularly polarized emissions observed by optical microscopy (Fig. 3e and Supplementary Fig. 21), with fluorescence intensity differences exceeding 30% under left- and right-handed polarizers (Fig. 3f). The results in Supplementary Fig. 22 indicated that the spin-photon design did not influence the fluorescence lifetime of the laser dyes. The chiroptical activity was then quantified by the strong signals in the circularly polarized luminescence spectra with large g_{lum} up to 0.41 (Fig. 3g,h, Supplementary Fig. 23 and Table S1). We also collected the temperature-dependent CPL and g_{lum} spectra, and conducted heating–cooling cycle tests (10-40 °C), during which the constant CPL and g_{lum} values of the spin-photon fibres indicated their fine temperature stability and reversibility (Supplementary Figs. 24 and 25). The 0.36 g_{lum} value of the fibres kept at 31.9 °C (average temperature of the hand) for 24 h, further demonstrating their satisfactory CPL properties during sustained wearable operation (Supplementary Fig. 26). Moreover, polarization analyzer measurements (Supplementary Fig. 27) showed the sinusoidal modulation response[18,30,31] when varying the rotation angle between the quarter-wave plate and linear

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polarizer (Fig. 3i and Supplementary Note 6), confirming high circular polarization purity. Polar plots further demonstrated low sensitivity to linear polarization angles (Fig. 3j), excluding contamination by linear polarization noise. Considering the follow-up practical applicability, we also tested its circular polarization characteristics under 274 lux ambient light (equivalent to bright office illumination), where the fibres maintained stable polarization discrimination as well, verifying readiness for practice (Fig. 3k). Additionally, the 8-meter-reach and omnidirectional transmission of a single fibre fully met interaction demands (Supplementary Fig. 28).

Polarization sensing

To evaluate our prepared spin-photon fibres in wearable sensing, we first wove them to form a glove. We adopted the computational imaging framework—integrating spin-photon emission from the glove as the polarization-selective signal source with real-time differential algorithms for image reconstruction (Fig. 4a, Supplementary Figs. 29-31 and Supplementary Note 7). This new polarization-enhanced sensing technology achieved exceptional performance (NSNR = 1.0, SNR = 82.04 dB), breaking traditional sensing limitations. Specifically, the generated photonic-spin signals were captured by left and right circularly polarized channels through a CMOS sensor array, transferred into analog voltage waveforms via photoelectric conversion processes, and then digitized by a 16-bit analog-to-digital converter (ADC). After that, the signal processor integrated and output these digital data, while the real-time differential algorithm isolated the spin-polarized components from interference by subtracting channel grayscale values, and finally mapped the intensity distribution into photonic-spin density images[21,32,33]. This

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developed technique enables high-fidelity photonic-spin signals visualization and interference-free sensing even in cluttered optical fields.

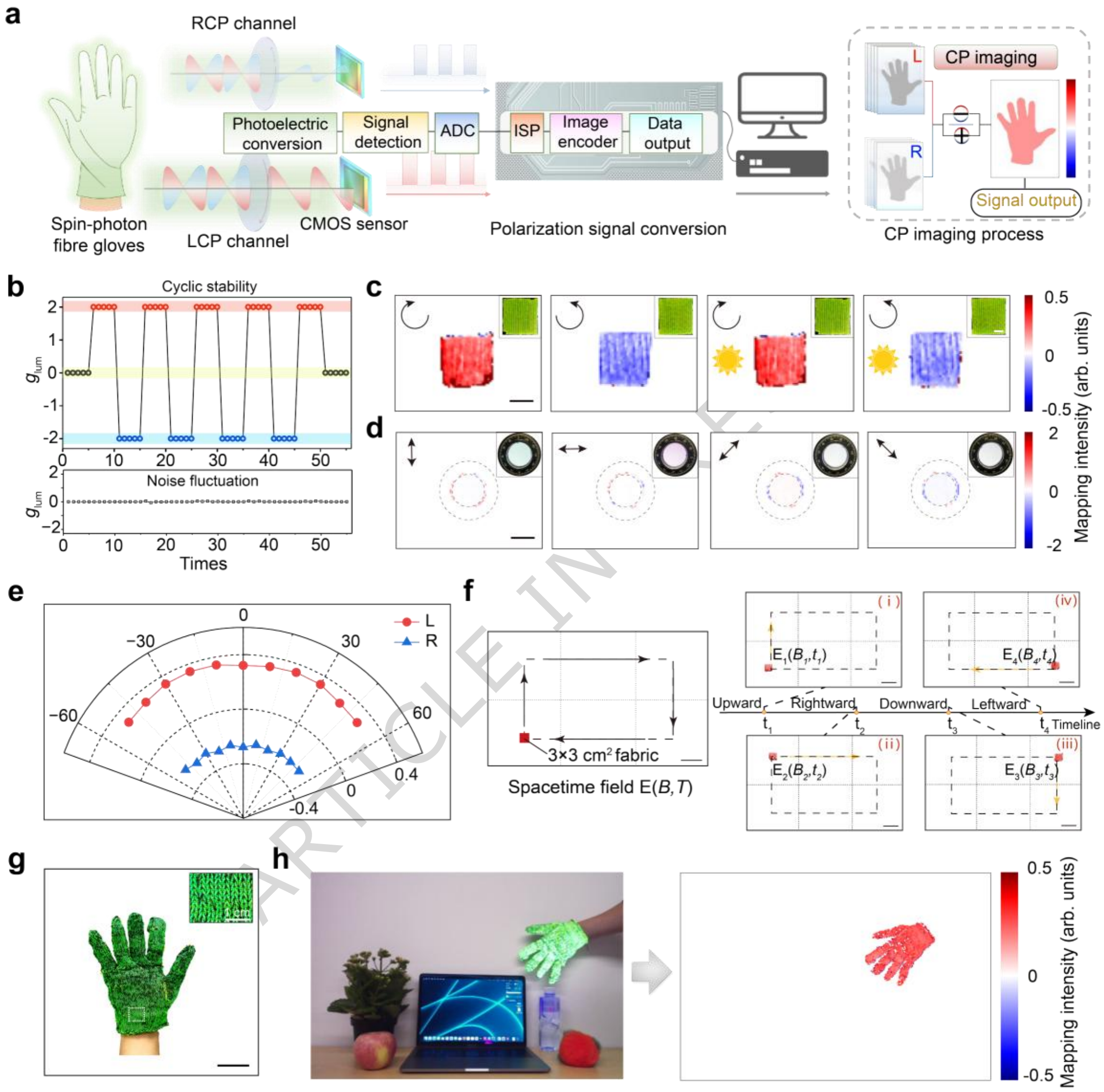

Fig. 4 | Polarization-enhanced sensing. a, System-level block diagram of photonic-spin signals decoding with a fibre-woven glove. Photonic-spin signals were first collected by a dual-channel signal acquisition, and then converted to digital signals. These signals were real-timely processed to achieve spin-density mapping. **b**, Testing of the loop stability of real-time imaging systems (top) and noise measurement (bottom) in bright scenes. **c**, Spin-imaging of left- and right-handed fibre-woven fabrics in dark field (left) and under 274 lux (right), demonstrating excellent anti-interference stability. Scale bar, 1 cm. **d**, Response spectra of imaging system to linear polarized inputs at 0°, 45°, 90°, and 135°. Scale bar, 1 cm. **e**, g_{lum} values of left- and right-handed fibres as a function of imaging angles. **f**, Dynamic decoding imaging of photonic-spin signals (under 274 lux). **g**, Photograph of a fibre-woven glove. Scale bar, 5 cm. **h**, Photograph (left) and spin density mapping image (right) of the circularly-polarized glove in complex light field.

For practice of our decoding system, we began with its characteristic calibration[34] , whose outstanding repeatability and ultralow dark polarization noise ensuring reliable spin-signal identification (Fig. 4b and Supplementary Fig. 32). We also synthesized non-CPL fibres to establish a baseline and measured the NSNR of the spin-photon fibres with and without baseline correction. The almost unchanged results (NSNR of 1 and 0.994) proved that the material fully enabled the improvement in anti-interference performance (Supplementary Fig. 33). Moreover, the calculated SNR of 82.04 dB, confirming the exceptionally low noise level of the system (the criterion for high-quality images is 40 dB, Supplementary Fig. 34). The photonic-spin imaging under 0 lux (darkness) and 274 lux (daylight simulation) further confirmed the system's performance (Fig. 4c and Supplementary Fig. 35). The negligible response to 0°, 45°, 90°, and 135° linearly polarized inputs was found, also validating the spin-signal specificity (Fig. 4d, Supplementary Figs. 36 and 37). We then quantified the observable angular range, achieving accurate photonic-spin signals detection at tilt angles up to ± 50° (Fig. 4e). The real-time mapping rate of 30 frames per second also ensured seamless integration with dynamic systems, such as wearable sensing robotic arms and VR (Fig. 4f and Supplementary Fig. 38).

The involved interactive glove showcased the wearability and conformability to the human body (Fig. 4g and Supplementary Fig. 39). The 60-washing-cycle in commercial washing machines implied its stability (Supplementary Figs. 40-42). We further examined the CPL and g_{lum} spectra of the spin-photon fibres under different wearable-relevant deformations such as finger flexion (stretching), local bending (bending) and torsion (twisting), and confirmed their considerable robustness in practical scenarios

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(Supplementary Figs. 43-45). Besides, its spin-emission remained accurately deciphered even under intense signal interference, enabling target signal decoupling from overlapping noise sources (Fig. 4h,i).

Polarization-enhanced sensing applications

A multimodal sensing platform (Fig. 5a) was then built, showing a NSNR of 1.0 (SNR = 82.04 dB) and a signal entropy reduction of 92.63% (Supplementary Fig. 46 and Supplementary Note 3), and thus significantly enhancing accuracy in machine vision tasks (*e.g.*, edge detection, semantic segmentation, and information recognition). Coupled with deep learning algorithms for gesture recognition and the Robot Operating System (ROS) programming framework, our platform could enable accurate sensing for VR and remote robotic arm precise control.

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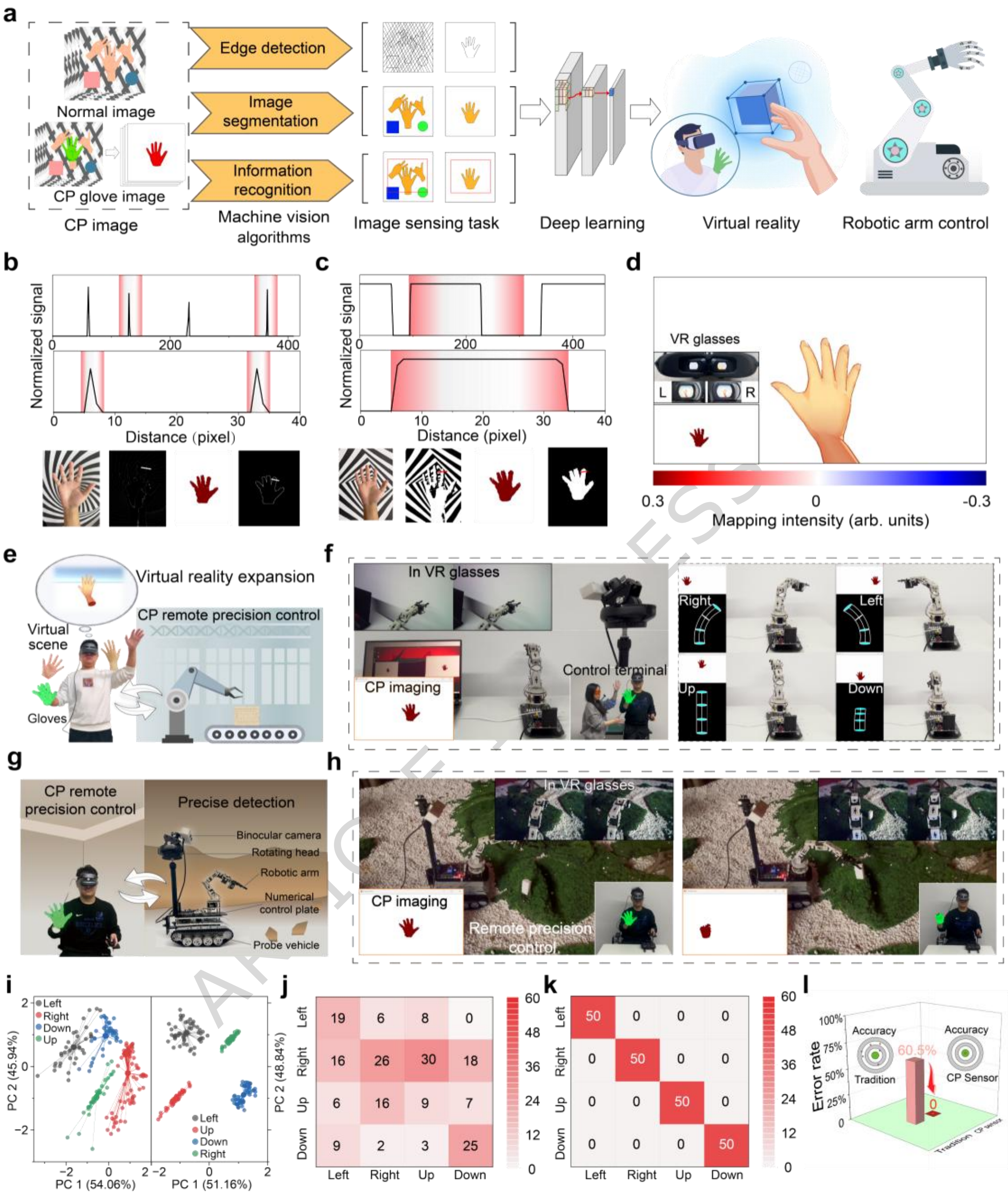

Fig. 5 | Wearable sensing applications. a, Diagram of the multi-modal sensing and applications based on fibre-emitted photonic-spin signals. Polarization-enhanced sensing involving signal generation and decoding, was applied to machine vision tasks, and then extended to virtual reality (VR) and robotic control. **b-c**, Normalized signal spectra of edge detection (**b**) and semantic segmentation performance (**c**) using polarization-enhanced (top) versus conventional imaging (bottom), with corresponding images. **d**, Images of VR interaction (mirrored) using fibre-woven glove. **e**, Conceptual image of the immersive precision operation interface with robotic arms utilizing polarization-enhanced sensing for target grasping. **f**, Polarization-enhanced sensing for remote precision control with robotic arm. **g-h**, Demonstration of Mars rover for target grasping based on polarization-enhanced sensing. **i**, Cluster analysis of robotic arm trajectory using

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conventional (left) and polarization-enhanced (right) sensing under multiple hand interference. **j-k**, Confusion matrix of target recognition using conventional sensing (**j**) and polarization-enhanced sensing (**k**). **l**, Error rate of the conventional (60.5%) and polarization-enhanced sensing (0%).

To practicalize it, we started with comparative studies of the polarization-enhanced and conventional sensing in cluttered environments, comparing their stark performance difference in machine vision tasks. For edge detection, polarization-enhanced sensing directly yielded clean, continuous contours without post-processing, whereas conventional methods required extensive noise filtering[17,32,35,36] (Fig. 5b and Supplementary Fig. 47). For semantic segmentation, the former exhibited sharp boundaries for finger regions leveraging the photonic-spin signals decoupling to bypass interference[37], while the other one failed (Fig. 5c and Supplementary Fig. 48). Additionally, polarization-enhanced sensing provided more precise outputs without irrelevant signals than the conventional technology in information recognition (Supplementary Fig. 49). These stemmed from the intrinsic ability of the involved photonic-spin signals to isolate target ones[21,23,38], enabling direct downstream algorithm integration for complex engineering applications[39].

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Building upon these, we engaged this technology in 100%-precision VR (Fig. 5d and Supplementary Fig. 50), and developed an immersive accurate robotic arm control system (Fig. 5e). The operator wearing the circularly-polarized glove, achieved interference-resistant manipulation while receiving real-time visual feedback via VR headsets (Fig. 5f and Supplementary Fig. 51). We further measured the core real-time metrics of the robotic arm teleoperation system, with the results proving its suitability in interaction demos (Supplementary Figs. 52 and 53). We then extended it to the remote Mars rover scenario (Fig. 5g and Supplementary Fig. 54), where polarization-enhanced guidance ensured 100% success in target retrieval within simulated Martian terrain (Fig. 5h and Supplementary Fig. 55). We performed cluster

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analysis on the robotic arm's motion trajectories (Fig. 5i)—controlled by our sensing significantly superior to conventional methods under multi-hand interference. This was further quantified by confusion matrices (Fig. 5j-l). The leap to 100%-precision originated from the immunity of the photonic-spin signals to environmental noise[15,21,23,40], establishing a paradigm for fault-intolerant applications in high-stakes fields.

In summary, we developed a discrete helix anchoring strategy—embedding spin-polarized states into unlimitedly-productive fibres—and achieved wearable polarization-enhanced sensing. Integration of the generation, decoupling, and transmission of photonic-spin signals into a single fibre enabled a theoretical-limit normalized signal-to-noise ratio of 1.0 and a signal entropy reduction of 92.63%. It also achieved error-free sensing in signal-coupled and interference-laden environment, and demonstrated multi-dimensional robotic control with 100% accuracy. The polarization-enhanced sensing and its applications in robotic arm teleoperation, VR, and machine vision, met the demand of environment-agnostic sensing in modern industries development. This work established a universal interface for next-generation intelligent textiles, heralding a new era of error-free sensing in human-machine interfaces.

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Methods

Materials

The nematic liquid crystal E7 and chiral dopants R/S5011 were sourced from Shijiazhuang Yesheng Chemical Technology Co., Ltd. Fluorescent dyes including 4-(dicyanomethylene)-2-methyl-6-(4-dimethylaminostyryl)-4H-pyran (DCM, 95% purity), coumarin 7 (C7, 98%) and coumarin 1 (C1, 98%) were procured from Sigma-Aldrich. Glycerol, calcium chloride, chloroform and n-hexane were obtained from Sinopharm Chemical Reagent Co., Ltd., while sodium alginate was acquired from Macklin Chemical Technology Co., Ltd. All chemical reagents were used without further purification as commercially supplied.

Preparation of spinning pulp

The cholesteric liquid crystal mixtures were formulated by doping chiral dopants S5011 and R5011 into nematic liquid crystal E7. The photonic bandgap was modulated by controlling dopant concentrations to achieve red (2.3 wt%), green (2.8 wt%), and blue (3.3 wt%) chiral structural colors. Corresponding fluorescent dyes—DCM (0.8 wt%, red emission), C7 (0.8 wt%, green), and C1 (1 wt%, blue)—were introduced into their respective photonic bandgap-tuned cholesteric systems. Chloroform was added for homogeneous dispersion, followed by 8-min sonication and solvent evaporation at 80 °C in an oven to obtain the composite mixtures.

A 3.5 wt% sodium alginate hydrogel precursor solution was prepared by gradual addition of sodium alginate to deionized water under controlled conditions to prevent flocculent aggregation. The beaker was maintained at 60 °C in a water bath with continuous magnetic stirring for 5 h until complete dissolution, yielding a homogeneous transparent solution. After cooling to ambient temperature, the solution was degassed for 5 min using a vacuum system to eliminate residual bubbles. Concurrently, a 4 wt% calcium chloride crosslinking solution was prepared for hydrogel curing.

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Microfluidic spinning process

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The inner-phase liquid crystal composite and outer-phase sodium alginate hydrogel precursor were delivered via independent microfluidic syringe pumps (models QHZS001A and YHBP01, Yuanhangdongli Technology Co., Ltd.) to the coaxial spinneret's core and shell channels, respectively. We have chosen coaxial spinneret nozzles with an inner diameter of 200 μm and an outer diameter of 900 μm. We have set the flow rate parameters in the microfluidic injection pump, with the inner phase flow rate at 0.58 $mm \cdot min^{-1}$ and the outer phase flow rate at 0.51 $mm \cdot min^{-1}$. Precise flow rate control established a biphasic fluidic system. Under shear stress within the needle, the liquid crystal phase underwent Rayleigh-Plateau instability, fragmenting into monodisperse microspheres that were encapsulated by the alginate solution to form core-shell precursor fibres. These fibres were subsequently vertically extruded into a $CaCl_2$ coagulation bath, where sufficient residence time ensured complete ionic crosslinking. Continuous fibre collection was achieved using a servo motor-driven spooling system, followed by rotational solvent exchange in a glycerol immersion bath.

Numerical simulation

To elucidate the microfluidic spinning mechanism, a phase-field numerical model was developed to simulate the coaxial flow within the spinneret, whereby the oil-phase liquid crystal core flow and aqueous-phase gel sheath flow were co-extruded into a cylindrical needle channel. Numerical solutions were obtained using COMSOL Multiphysics 6.0 software (COMSOL Inc.), enabling simulation of the dynamic interfacial shear processes and visualization of associated flow field profiles. The complete mathematical formulation and boundary conditions are provided in the Supplementary Information.

Spin-photon textile fabrication and processing

Spin-photon textiles were fabricated using a manual loom (Huanxibeier) through continuous weaving with single-strand spin-photon fibres filaments. To validate co-spinning compatibility with commercial fibres, hybrid textiles were produced by integrating cotton and polyurethane yarns with spin-photon fibres filaments on conventional weaving equipment. For embroidery processing, an industrial embroidery machine (Feiren) equipped with spin-photon fibres threads was employed. Prior to embroidery, critical

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parameters including needle penetration frequency (2 Hz) and stitch spacing (0.8 mm) were systematically adjusted to ensure substrate compatibility with commercial fabrics.

Bending radius quantification

Optical fibres were affixed onto glass slides using adhesive tape and mounted on an optical microscope stage. The objective lens magnification was adjusted to fully encompass the bending region within the field of view, with transmitted light illumination optimized to enhance edge contrast. Multi-focal plane images were captured, from which the frame exhibiting optimal fibre curvature definition was selected for analysis. The digitized image was processed using ImageJ or Adobe Illustrator software, wherein the fibre's bending profile was traced using freehand selection or brush tools to delineate the central axis. An inscribed circle was computationally generated along the curvature, enabling measurement of the curvature radius. Absolute dimensions were determined through scale calibration, with final values derived from averaging repeated measurements.

Laundering durability assessment

Three designated measurement zones were marked on the finger regions of spin-photon fibres knitted gloves for baseline characterization (0 laundering cycles). The gloves were subjected to accelerated laundering in a commercial top-loading washer (Shishang) under domestic laundering conditions: water temperature maintained at 25 °C, 10 min per cycle, with standard detergent and companion fabric loading to simulate routine washing. Sequential laundering cycles (20, 40, 60 cycles) were performed with intermediate fluorescence characterization. Immediately following each target cycle count, samples were retrieved for optical evaluation. Fluorescence intensity measurements were conducted at all three fingertip zones, with cycle-dependent performance calculated through spatial averaging.

Polarization characterization

To investigate the circular polarization properties of spin-photon fibres, a custom optical characterization platform was established. A collimated 365 nm ultraviolet light source was aligned with the optical axis to ensure beam-path consistency. The UV beam irradiated fibres samples mounted on a stage, inducing

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wavelength-specific responses. The emitted light was subsequently modulated through sequential polarization elements: a linear polarizer (400 to 700 nm, Thorlabs) followed by a quarter-wave plate (QWP, 350 to 850 nm, Thorlabs) for polarization state decomposition. The processed optical signals were coupled into a fibre-optic spectrometer via an optical fibre assembly. Intensity measurements were recorded as functions of polarizer/waveplate angular orientations (0-360° in 15° increments), with sinusoidal periodicity analysis enabling precise determination of circular polarization fluctuating characteristics. All measurements were performed under darkroom conditions (ambient light < 1 lux) to minimize photonic interference.

Machine vision task evaluation

To evaluate the performance of polarization-enhanced sensing versus conventional imaging in edge detection and image segmentation tasks, image datasets from both sensing modalities were acquired and processed using ImageJ software. For edge detection, the workflow comprised: grayscale conversion of raw images, intensity inversion to reduce background signals, Gaussian blur denoising, adaptive threshold binarization, and edge extraction using the built-in Sobel operator. The image segmentation protocol followed identical grayscale conversion, inversion, and Gaussian denoising steps, succeeded by automated threshold binarization and morphological operations (hole filling, erosion, and dilation) to optimize segmentation accuracy. Full methodological details are provided in the Supplementary Information.

Virtual reality integration system

The GOOVIS Lite VR head-mounted display was employed to deliver VR visual stimuli, featuring dual micro-OLED displays with 4,496 pixels per inch (PPI) resolution. The stereoscopic VR environment was rendered through a Unity 3D platform running on a host workstation, with wireless data transmission facilitated by a wireless transmitter (Goovis Tech). Real-time video streams acquired through both polarization-enhanced and conventional imaging modalities were processed by a hand-tracking algorithm that extracted skeletal keypoints at 30 fps sampling rate. These kinematic data packets were subsequently fed into the Unity3D rendering engine via UDP protocol for virtual-object matching and real-time scene

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updating, enabling immersive interaction through closed-loop visuomotor synchronization. Complete system schematics is provided in Supplementary Information.

Robotic arm control system

The robotic arm (Yahboom Intelligent Technology Co., Ltd.) was interfaced with a PC-hosted virtual machine serving as the central controller. Real-time motion control was achieved through command transmission from the host computer. A custom C++ program was developed to integrate polarization-enhanced sensing modules, implementing sequential processes: hand-position data acquisition, PID control implementation, serial communication protocols, and inverse kinematics calculations. Processed motion vectors were ultimately transmitted to servo motor drivers via modulation signals. Complete system architecture diagrams and control algorithms are detailed in the Supplementary Information.

Characterization

Optical properties were systematically characterized using the following instrumentation: UV-vis-NIR transmission spectra were acquired with a Shimadzu 3700 DUV spectrophotometer, with fluorescence spectra acquired using a Hitachi F-4700 spectrofluorometer. Chiroptical measurements included circular dichroism (CD) spectra recorded on a JASCO J-1500 spectropolarimeter and circularly polarized luminescence (CPL) spectra obtained via a JASCO CPL-300 system. Microstructural analyses were performed using an upright polarized optical microscope (Mshot MP41) for bright-field and POM imaging, complemented by Fourier-transform infrared spectroscopy (FTIR) measurements conducted on a Nicolet iN10MX FTIR microscope. Surface morphology was examined using a field-emission scanning electron microscope (Zeiss Supra 40, Carl Zeiss AG) operated at 1.5 kV acceleration voltage.

Data availability

The dataset supporting the findings of this study has been deposited in the Zenodo repository[41], and is available at https://doi.org/10.5281/zenodo.20742889. Source data are provided with this paper.

Code availability

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The code that supports the findings of this study is available on GitHub: https://github.com/Zhuang-Lab6/NCOMMS-25-98213.git.

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Author contributions

T.Z. and G.L. conceived the idea and designed the research. T.Z., G.L. and Y.Z. wrote the paper. G.L. carried out the experiments and analyzed the results. G.L. and Y.Z. helped to prepare figures. Y.W., Q.G., S.Z., M.Z., J.L., Z.L., Y.H. and A.L. helped to collect the data. All authors discussed the results and assisted during manuscript preparation.

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Competing interests

The authors declare no competing interests.